Inference for Multiple Linear Regression

Checking for any linear relationship

Consider the restaurant data again:

y: dinner price

x_1: food quality score, on 0-to-30 scale

x_2: service quality score, on 0-to-30 scale

Fit a linear regression model using food quality and service quality as predictors.

```
combined_model<-lm(Price ~ Food + Service,
                   data=restaurants)
```

Checking for any linear relationship

Goal: test the hypothesis

$$H_0 : \beta_1 = \beta_2 = 0$$

against the alternative hypothesis that at least one of β_1, β_2 is not zero.

```r
summary(combined_model)
```

```
##
## Call:
## lm(formula = Price ~ Food + Service, data = restaurants)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16.1333  -4.7053   0.4169   3.5992  27.0728
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -21.1586     5.6651  -3.735 0.000258 ***
## Food          1.4954     0.4462   3.351 0.000997 ***
## Service       1.7041     0.4185   4.072 7.22e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.942 on 165 degrees of freedom
## Multiple R-squared:  0.4486, Adjusted R-squared:  0.4419
## F-statistic: 67.12 on 2 and 165 DF,  p-value: < 2.2e-16
```

Reading the output

```
Residual standard error: 6.942 on 165 degrees of freedom
Multiple R-squared:  0.4486,    Adjusted R-squared:  0.4419
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```

The bottom line of the summary output gives the value of the test statistic, $f_{obs} = 67.12$, and the corresponding p value.

The notation "p-value $<$ 2.2e-16" means that the p value is smaller than 2.2×10^{-16}, i.e. 0.00000000000000022

Test conclusion

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```

Ignoring "food" and "service" as predictors, results in significantly more unexplained variation in the price values.

Reject H_0. Conclude at least one of food quality and service quality has a linear relationship with price (test doesn't say which one (or both)).

Adding a predictor

Next, consider an expanded model that also includes the restaurant's decor quality as a predictor (scored on a 0 to 30 scale)

```
Call:
lm(formula = Price ~ Food+Service+Decor, data=restaurants)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -24.6409     4.7536  -5.184 6.33e-07 ***
Food          1.5556     0.3731   4.170 4.93e-05 ***
Service       0.1350     0.3957   0.341    0.733
Decor         1.8473     0.2176   8.491 1.17e-14 ***
---

Residual standard error: 5.803 on 164 degrees of freedom
Multiple R-squared:  0.617, Adjusted R-squared:   0.61
F-statistic: 88.06 on 3 and 164 DF,  p-value: < 2.2e-16
```

Test about subset of predictors

In this expanded model, we could also test a hypothesis about a subset of the predictors, like

$$H_0 : \beta_2 = \beta_3 = 0$$

In other words, check whether "reduced" model is significantly worse than "full" model.

full: $\text{price} = \beta_0 + \beta_1 \text{food} + \beta_2 \text{service} + \beta_3 \text{decor} + \varepsilon$

reduced: $\text{price} = \beta_0 + \beta_1 \text{food} + \varepsilon$

Test about subset of predictors

Fit the "reduced" model using only food quality:

```
reduced_model<-lm(Price ~ Food, data=restaurants)
```

Fit the "full" model using all three predictors:

```
full_model<-lm(Price~ Food + Service + Decor,
               data=restaurants)
```

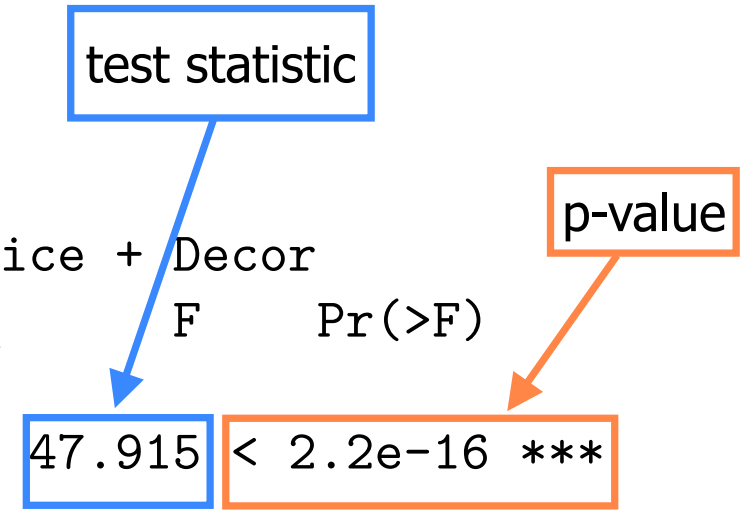
Compare models using anova

To compare them, use the anova() function.

If we specify the reduced model and full model that we want to compare, the anova() function performs the relevant F test of our hypothesis.

```
anova(reduced_model, full_model)
```

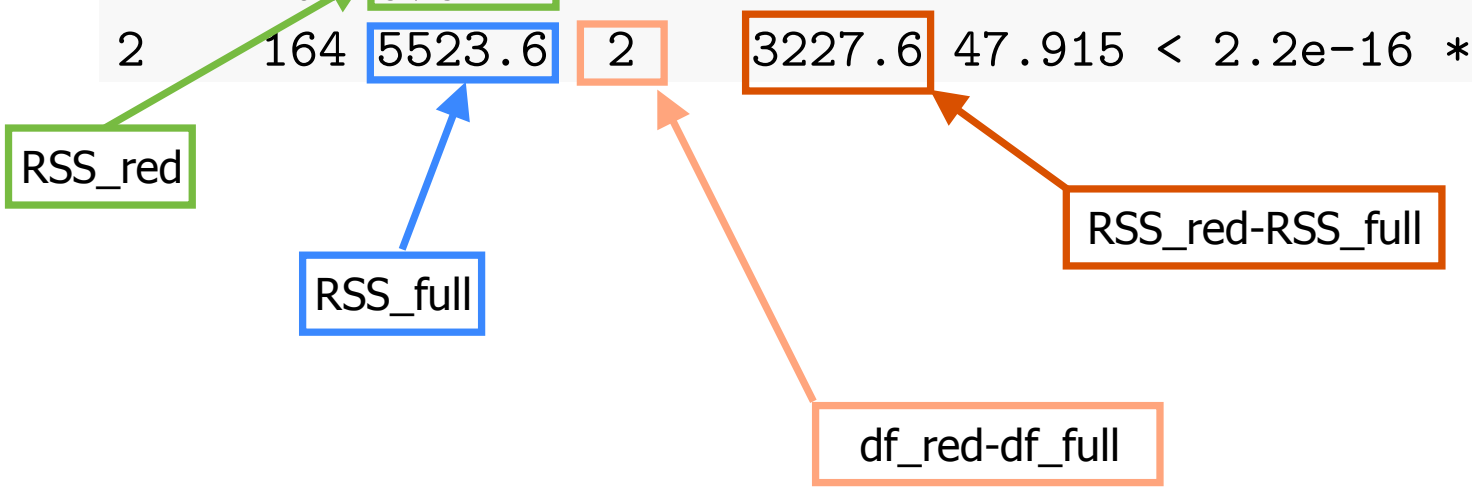
```
## Analysis of Variance Table
##
## Model 1: Price ~ Food
## Model 2: Price ~ Food + Service + Decor
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1    166 8751.2
## 2    164 5523.6  2    3227.6 47.915 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' '
```



Reading the output

```
Analysis of Variance Table

Model 1: Price ~ Food
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Res.Df    RSS Df Sum of Sq      F    Pr(>F)
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Test conclusion

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Analysis of Variance Table

Model 1: Price ~ Food
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1    166 8751.2
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```

p-value is extremely close to zero

Reject $H_0 : \beta_2 = \beta_3 = 0$, and conclude that at least one of the variables "service" and "decor" DOES provide additional information about restaurant prices, *in the context of a linear model that also uses food quality as a predictor*.